Detection of Near and Far Objects Using Depth Anything

In this part, the **Depth8 Dataset** was used to analyze the depth of objects at different distances. The selected image was **image_08.png**, which contains objects located at different depths. The RGB image was processed using **Depth Anything V2 Small** to generate a relative depth map.

Three regions of interest (ROIs) were manually selected:

- **Object A (Near):** An object close to the camera
- **Object B (Medium):** An object at an intermediate distance
- **Object C (Far):** A distant object/background region

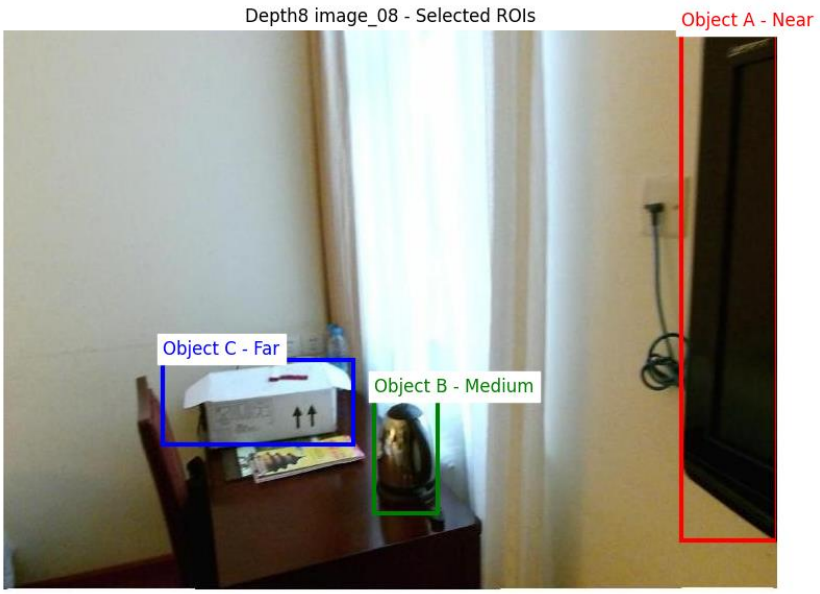


Figure 1. Selected near, medium, and far objects and their corresponding ROIs in the image.

For each ROI, the mean, median, minimum, and maximum depth values were calculated from the predicted depth map.

Depth Anything V2 Results (Relative Depth)

Object	Mean Depth	Median Depth	Minimum Depth	Maximum Depth
Object A (Near)	6.636	6.738	4.341	8.296
Object B (Medium)	2.328	2.549	0.886	2.809
Object C (Far)	1.340	1.521	0.472	2.989

The results show that the predicted depth values correctly preserve the relative ordering of the objects. In this experiment, larger predicted depth values correspond to closer objects, indicating that the model output behaves as an inverse-depth-like representation.

Ground Truth Verification

Since the Depth8 dataset provides metric depth maps, the same ROIs were also evaluated using the ground-truth depth values. The ground-truth depth represents the actual distance in meters.

Object	Mean GT Depth (m)	Median GT Depth (m)	Minimum GT Depth (m)	Maximum GT Depth (m)
Object A (Near)	0.751	0.755	0.561	1.049
Object B (Medium)	1.473	1.393	1.236	2.319
Object C (Far)	1.842	1.709	1.451	2.341

The ground-truth values confirm the expected distance relationship:

Object A (Near) < Object B (Medium) < Object C (Far)

in terms of physical distance.

The Depth Anything V2 predictions show the same ordering but with an opposite numerical trend because the model provides **relative depth**, not metric depth.

Analysis Question

If the model produces depth values of **0.21** and **0.81** for two objects, it cannot be concluded that the objects are located at **2 meters** and **8 meters**.

Depth Anything V2 does not directly predict physical distance. Its output represents relative depth information, where the values indicate the ordering of objects in the scene rather than their actual distances. To obtain metric depth in meters, the model output must be calibrated using known depth measurements or ground-truth depth data.

Therefore, Relative Depth ≠ Metric Depth and the predicted values cannot be directly interpreted as real-world distances without additional calibration.

Appendix

The difference between **mean** and **median** is how they summarize the values inside an ROI (region of interest).

Mean (Average)

The **mean** is the normal average of all pixel depth values.

The mean is affected strongly by extreme values (outliers).

Median

The **median** is the middle value after sorting the data:

Sorted values:

1.0, 1.2, 1.3, 1.5, 5.0

The middle value (Median) is 1.3

The median is less affected by extreme values.

In depth analysis

For a selected object:

- **Mean depth** tells the average predicted depth of all pixels in the ROI.
- **Median depth** tells the typical depth of the object and is more robust to noise, edges, and background pixels.

For example, if an ROI contains mostly an object but some background pixels:

Object pixels: 2.0, 2.1, 2.2, 2.3

Background pixels:6.0

Mean: (2.0+2.1+2.2+2.3+6.0) / 5 = 2.92

Median: 2.2

The median better represents the actual object depth.

For Object A:

Mean Depth: 6.636, Median Depth: 6.738

The values are close, meaning the ROI has a relatively consistent depth.

For Object C:

Mean Depth: 1.340, Median Depth: 1.521

The difference is larger, suggesting more depth variation inside the ROI (for example, object boundaries, background pixels, or depth changes).

In depth estimation tasks, **median depth is often preferred for representing the distance of an object**, while **mean depth is useful for describing the overall depth distribution inside the ROI**.